

6.1 Deformation: Stress & Strain

Question Paper

Course	CIE A Level Physics
Section	6. Deformation of Solids
Topic	6.1 Deformation: Stress & Strain
Difficulty	Hard

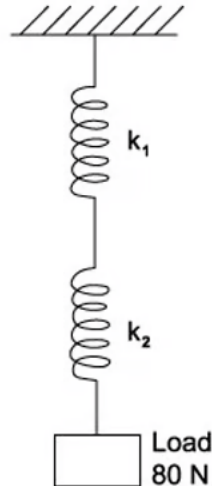
Time allowed: 20

Score: /10

Percentage: /100

Question 1

A student connected two springs, one with spring constant $k_1 = 4 \text{ kN m}^{-1}$ and the other with spring constant $k_2 = 2 \text{ kN m}^{-1}$ as shown in the diagram below.



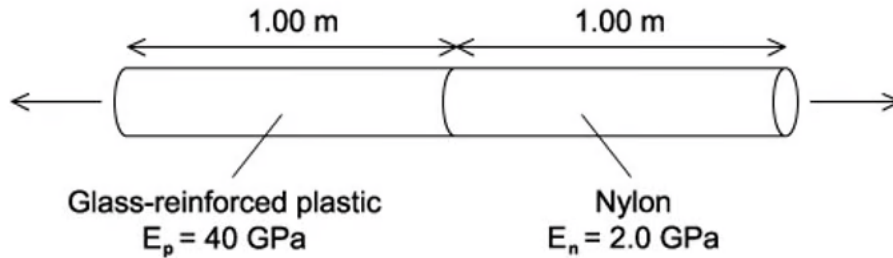
When a load of 80 N is applied, what is the total extension?

- A. 60 cm
- B. 6 cm
- C. 4 cm
- D. 1.3 cm

[1 mark]

Question 2

A glass-reinforced plastic rod and a nylon rod attached, as shown, was used to make a composite rod.



Each rod has a length of 1.00 m and the same cross-sectional area. The Young modulus E_n of the nylon is 2.0 GPa, and the Young modulus E_p of the plastic is 40 GPa

The composite rod will break when its total extension reaches 3.0 mm.

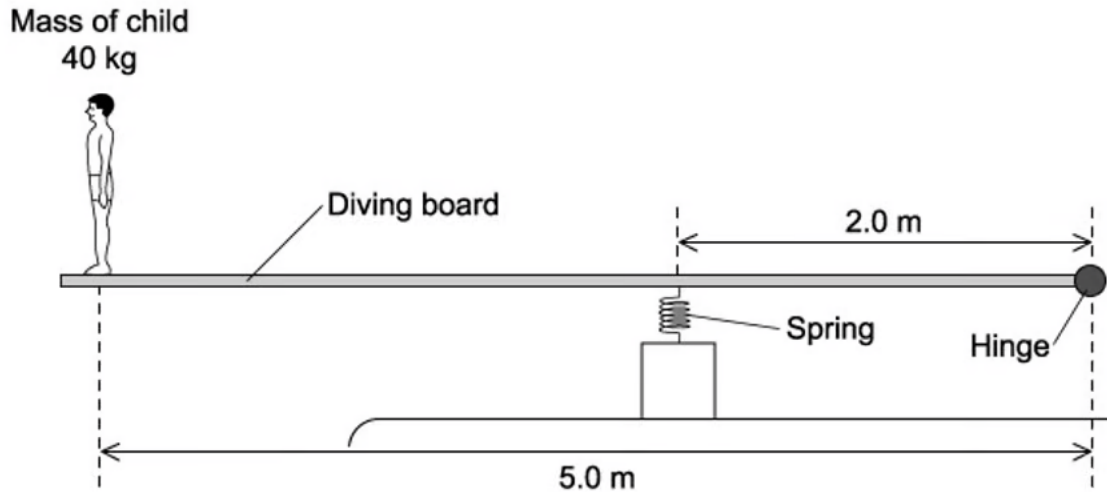
What is the greatest tensile stress that can be applied to the composite rod before it breaks?

- A. $5.7 \times 10^6 \text{ Pa}$
- B. $5.7 \times 10^9 \text{ Pa}$
- C. $7.1 \times 10^{-14} \text{ Pa}$
- D. $7.1 \times 10^{-2} \text{ Pa}$

[1 mark]

Question 3

A child with a mass of 40 kg was standing on a diving board. The diving board has a length of 5.0 m and is hinged at one end then supported by a spring 2.0 m from the hinged end. The spring has a spring constant of 10 kN m^{-1}



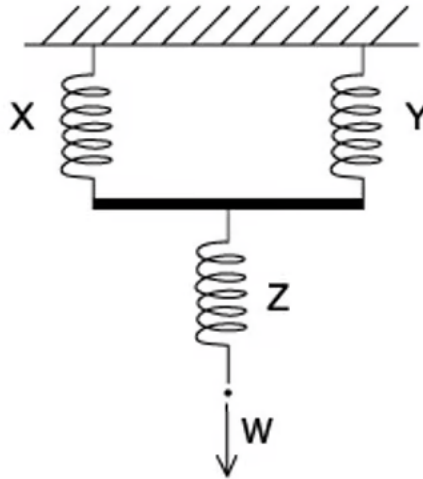
What is the extra compression of the spring caused by the child standing on the end of the board?

- A. 16 cm
- B. 9.8 cm
- C. 1.6 cm
- D. 1.0 cm

[1 mark]

Question 4

Three springs are arranged vertically, as shown.



Springs X and Y are identical and have a spring constant k . Spring Z has a spring constant of $3k$.

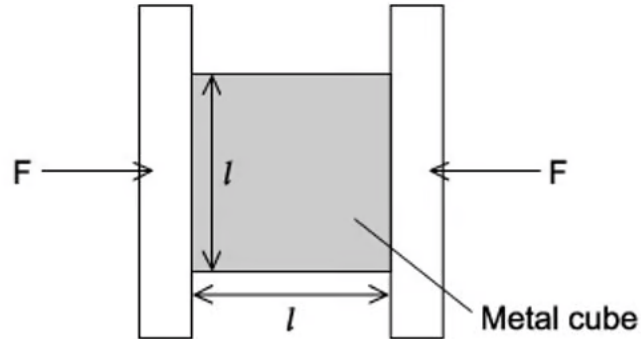
What is the increase in the overall length of the arrangement when a force W is applied as shown?

- A. $\frac{4W}{3k}$
- B. $\frac{5W}{6k}$
- C. $4kW$
- D. $\frac{7kW}{2}$

[1 mark]

Question 5

The diagram shows a metal cube is placed between two boards and compressed elastically by two opposing forces F . The cube has a length of each side of l



How will Δl , the amount of compression, relate to l ?

- A. $\Delta l \propto l$
- B. $\Delta l \propto l^2$
- C. $\Delta l \propto \frac{1}{l^2}$
- D. $\Delta l \propto \frac{1}{l}$

[1 mark]

Question 6

A metal wire that is supported vertically from a fixed point has a load of 84 N applied to the lower end. The wire has a cross-sectional area of 0.02 mm^2 and obeys Hooke's law. The length of the wire increases by 0.30 %.

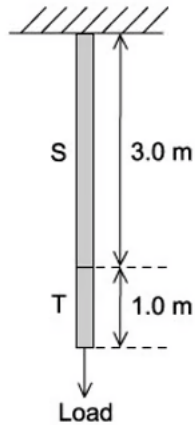
What is the Young modulus of the metal wire?

- A. $1.4 \times 10^{12} \text{ Pa}$
- B. $1.4 \times 10^9 \text{ Pa}$
- C. $1.4 \times 10^8 \text{ Pa}$
- D. $1.4 \times 10^5 \text{ Pa}$

[1 mark]

Question 7

Two wires S and T is joined, S has a length of 3.0 m and T a length of 1.0 m, both wires have a uniform cross-sectional area.



When a load was hung from the wires S stretched by 1.5 mm and T by 1.0 mm.

The same load is then hung from a second wire of the same cross-section, consisting of 1.0 m of metal S and 3.0 m of metal T.

What is the total extension of this second wire?

- A. 3.5 mm
- B. 2.5 mm
- C. 5.0 mm
- D. 4.8 mm

[1 mark]

Question 8

Two steel wires X and Y that both obey Hooke's law were used in an experiment. Wire X has a length l and a cross-sectional area A . Wire Y has a length of $2l$, and cross-sectional area of $\frac{A}{2}$.

What is the ratio $\frac{\text{tension in X}}{\text{tension in Y}}$ when both wires are stretched to the same extension?

A. $\frac{1}{4}$

B. $\frac{4}{1}$

C. $\frac{2}{1}$

D. $\frac{1}{2}$

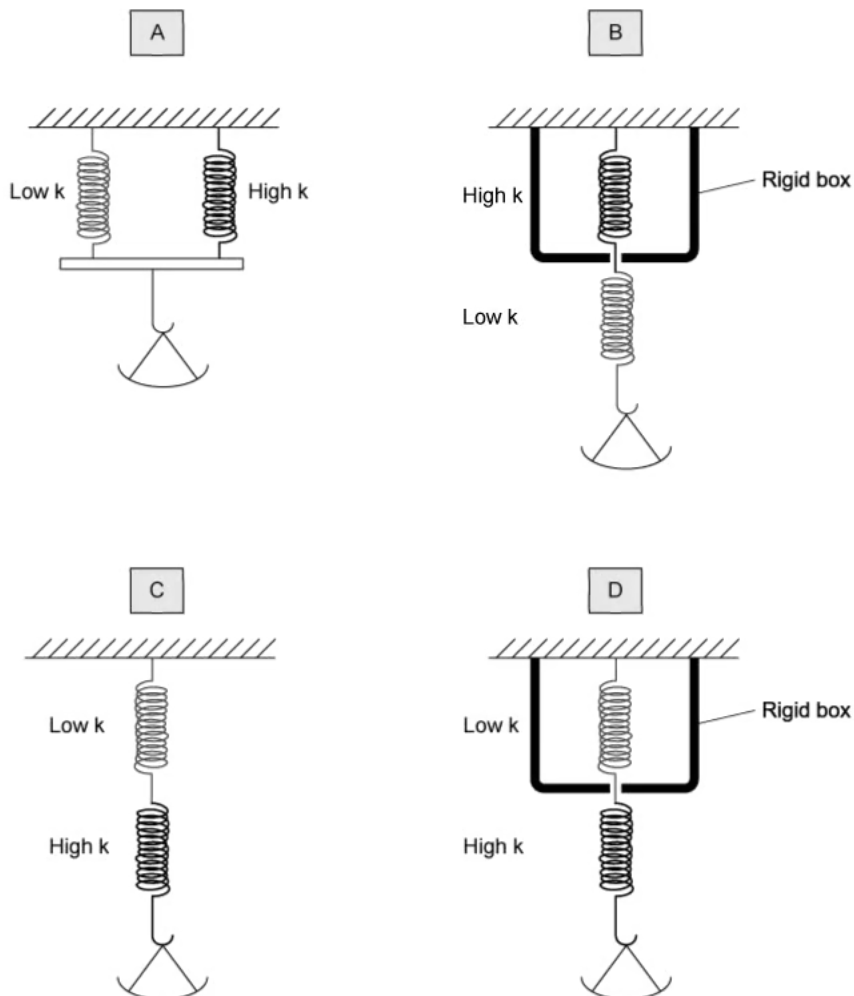
[1 mark]

Question 9

In order to measure the mass of flour a set of scales needs to have high sensitivity for small masses and a low sensitivity with larger masses.

To achieve this two springs are used with different spring constants k . One has a lower spring constant than the other.

Which of the following sets of springs would be suitable for this purpose?

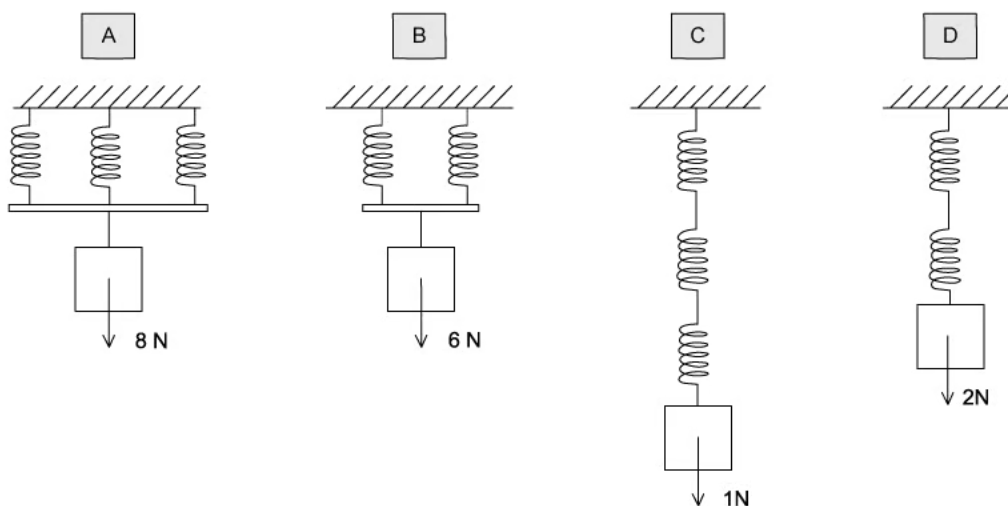


[1 mark]

Question 10

The springs in the diagram all have the same spring constant. They were arranged in a different formation with a different load, as shown in the diagram.

Which arrangement has the largest extension?



[1 mark]

Model Answers: Hard

1

The correct answer is **B** because:

- The spring constant is a measure of the stiffness of a spring, up to the limit of proportionality
- This system has two springs in series, each spring having a different spring constant
- For springs in series the effective spring constant k_{eff}
 - $\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$
 - $k_{eff} = \left[\frac{1}{k_1} + \frac{1}{k_2} \right]^{-1} = \left[\frac{1}{4} + \frac{1}{2} \right]^{-1} = \frac{4}{3} kN m^{-1}$
 - $k_{eff} = \frac{4}{3} kN m^{-1} = \frac{4000}{3} N m^{-1}$
- Hooke's law $F = kx$
 - Extension $x = \frac{F}{k_{eff}} = \frac{80}{4000/3} = 0.06 m = 6 cm$

2

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- Tensile stress is the force applied per cross-sectional area
- In both materials the same force is applied, the materials have different extensions for this force
- The total extension can't be equal to 3.0 mm as the rod would then have broken

- $E_p = \sigma \times \frac{1.0}{(40 \times 10^9)} = \frac{\sigma}{(40 \times 10^9)}$
- $E_n = \sigma \times \frac{1.0}{(2 \times 10^9)} = \frac{\sigma}{(2 \times 10^9)}$
- The greatest stress that can be added is the sum of the stress of both materials, and this should be equal to 3.0 mm
 - $E_n + E_p = 3.0 \times 10^{-3}$
 - $\frac{\sigma}{(2 \times 10^9)} + \frac{\sigma}{(40 \times 10^9)} = 3.0 \times 10^{-3}$
 - $\sigma \times \left(\frac{1}{(2 \times 10^9)} + \frac{1}{(40 \times 10^9)} \right) = 3.0 \times 10^{-3}$
 - $\sigma = 3.0 \times 10^{-3} / \left(\frac{1}{(2 \times 10^9)} + \frac{1}{(40 \times 10^9)} \right) = 5.71 \times 10^6 \text{ Pa}$
- Therefore, the greatest tensile stress that can be applied is $5.7 \times 10^6 \text{ Pa}$

3

The correct answer is **B** because:

- Hooke's law states $F = ke$, where k is the spring constant
- The child and the spring will cause a moment at a distance from the hinge
- At equilibrium, the torque of the child (anticlockwise) should be equal to the torque of the spring (clockwise)
- Let the force of the spring = F
 - Clockwise moments = $mgx = 40 \times 9.81 \times 5.0$
 - Anticlockwise moments = $Fd = F \times 2.0$
 - $40 \times 9.81 \times 5.0 = F \times 2.0$
 - $F = \frac{40 \times 9.81 \times 5}{2} = 981 \text{ N}$
- So, the spring force is equal to $F = 981 \text{ N}$
- Spring constant, $k = 10 \text{ kN m}^{-1} = 10000 \text{ N m}^{-1}$
 - Extension = $\frac{F}{k} = \frac{981}{10000} = 0.0981 \text{ m}$
- Therefore, the compression of the spring caused by the child is 9.8 cm

4

The correct answer is **B** because:

- Hooke's law states that $F = kx$ where k is the spring constant
- For springs in series the effective spring constant k_{eff}
 - $\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$
- For springs in parallel the effective spring constant
 - $k_{eff} = k_1 + k_2 + \dots$
- Springs X and Y are in parallel each with the same spring constant
 - $k_{eff} = k + k = 2k$
- The effective spring constant of the whole system, spring Z has a spring constant of $3k$, so:
 - $\frac{1}{k_{eff}} = \frac{1}{2k} + \frac{1}{3k}$
 - $k_{eff} = \left[\frac{1}{2k} + \frac{1}{3k} \right]^{-1} = \frac{6k}{5}$
- Therefore, extension is equal to $x = \frac{W}{k_{eff}} = W \times \left(\frac{5}{6k} \right) = \frac{5W}{6k}$

5

The correct answer is **D** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- Therefore, Young modulus $E = \frac{(F/A)}{(e/L)} = \frac{FL}{Ae}$
- Where:
 - Area $A = l^2$
 - Compression $e = \Delta l$
- So, Young modulus is proportional to:
 - $E \propto \frac{l}{l^2 \times \Delta l}$

- $E \propto \frac{1}{l \times \Delta l}$
- So $\Delta l \propto \frac{1}{El}$, giving $\Delta l \propto \frac{1}{l}$ as the Young modulus is constant

6

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- Where:
 - Force $F = 84 \text{ N}$
 - Area $A = 0.02 \text{ mm}^2 = 0.02 (\times 10^{-3}) \text{ m}^2 = 0.02 \times 10^{-6} \text{ m}^2$
- The wire increases in length by 0.3 %, this is equal to the strain, so:
 - $\text{Strain} = 0.3\% = \frac{0.3}{100} = 0.0030$
- Young modulus = stress / strain
 - $E = \frac{84}{(0.02 \times 10^{-6}) \times 0.003} = 1.4 \times 10^{12} \text{ Pa}$
- Therefore, the Young modulus of the metal wire is $1.4 \times 10^{12} \text{ Pa}$

7

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$

- $E \propto \frac{1}{l \times \Delta l}$
- So $\Delta l \propto \frac{1}{El}$, giving $\Delta l \propto \frac{1}{l}$ as the Young modulus is constant

6

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- Where:
 - Force $F = 84 \text{ N}$
 - Area $A = 0.02 \text{ mm}^2 = 0.02 (\times 10^{-3}) \text{ m}^2 = 0.02 \times 10^{-6} \text{ m}^2$
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7

The correct answer is **A** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- In a material the Young modulus remains constant
- As the load and cross-sectional area are the same, the stress $\left(\frac{\text{force}}{\text{area}} \right)$ is going to

be the same

- Metal S

- 1st experiment, $E = \frac{3}{1.5}$
- 2nd experiment $E = \frac{1}{x}$
- Therefore, $\frac{3}{1.5} = \frac{\text{stress}}{x}$
- Extension, $x = \frac{1.5}{3} = 0.5 \text{ mm}$

- Metal T

- 1st experiment, $E = \frac{1}{1}$
- 2nd experiment $E = \frac{3}{x}$
- Therefore, stress is equal to $= \frac{3}{x}$
- So, extension $x = 3.0 \text{ mm}$

- Total extension $0.5 + 3.0 = 3.5 \text{ mm}$

8

The correct answer is **B** because:

- The Young modulus is given by $E = \frac{\text{stress}}{\text{strain}} = \frac{\sigma}{\epsilon}$
 - Stress is equal to $\sigma = \frac{\text{force}}{\text{area}} = \frac{F}{A}$
 - Strain is equal to $\epsilon = \frac{\text{extension}}{\text{original length}} = \frac{\Delta L}{L}$
- As both wires are made of steel they have the same Young modulus, E
- Young modulus $E = \frac{(F/A)}{(x/L)} = \frac{FL}{Ax}$
- So, tension is equal to $F = \frac{Eax}{l}$
 - For wire X: $F_X = \frac{Eax}{l}$

$$\circ \text{ For wire Y: } F_Y = \frac{E(a/2)x}{2l} = \frac{Eax}{4l} = \frac{F_X}{4}$$

$$\bullet \text{ So, the ratio } = \frac{F_X}{F_Y} = \frac{F_X}{(F_X/4)} = 4$$

9

The correct answer is **D** because:

- The question states that a high sensitivity for small masses and a low sensitivity for larger masses is needed
- For springs in series the effective spring constant k_{eff}
 - $\circ \frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$
- For springs in parallel the effective spring constant k_{eff}
 - $\circ k_{eff} = k_1 + k_2 + \dots$
- Hooke's law states $F = kx$, where k is the spring constant
 - $\circ$ So, extension is equal to $x = \frac{F}{k}$
- Force $F = \text{weight} = mg$
 - $\circ$ For a small mass a higher extension, x is needed, so the spring constant k needs to be small
 - $\circ$ For a larger mass a lower extension, x is needed, so the spring constant k needs to be larger
- The spring with the lower spring constant is placed into a rigid box to reduce the sensitivity for larger masses

A is incorrect because when the springs are in parallel, they will extend different lengths giving incorrect measurements

B is incorrect because the spring with the higher spring constant inside the box would give high sensitivity at larger masses

C is incorrect because the springs in series without the box would give the same sensitivity for small and larger masses

10

The correct answer is **D** because:

- Hooke's law states $F = kx$, where k is the spring constant
- For springs in series the effective spring constant k_{eff}
 - $\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \dots$
- For springs in parallel the effective spring constant k_{eff}
 - $k_{eff} = k_1 + k_2 + \dots$
- Extension $x = \frac{F}{k_{eff}}$
- When the springs are added in parallel the effective spring constant is greater, so the extension is shorter

A	B
$k_{eff} = k_1 + k_2 + k_3 = 3k$ $x = \frac{8}{3k}$	$k_{eff} = k_1 + k_2 = 2k$ $x = \frac{6}{2k} = \frac{3}{k}$
C	D
$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} + \frac{1}{k_3} = \frac{3}{k}$ $x = 1 \times \frac{3}{k} = \frac{3}{k}$	$\frac{1}{k_{eff}} = \frac{1}{k_1} + \frac{1}{k_2} = \frac{2}{k}$ $x = 2 \times \frac{2}{k} = \frac{4}{k}$

- Therefore, option **D** has the largest extension